

Tutorial on Wearability and Usability Evaluation

ELEC-E7840 Smart Wearables Spring 2026

Agenda

- Design guidelines for wearability
- WEAR Scale
- Comfort Rating Scale
- System Usability Scale
- Tips for user tests

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Wearability

- **Wear-ability**, the physical shape of wearables and their active relationship with the human form
- Selected design guidelines for wearability (Gemperle et al. 1998)
 - 1) Design for dynamic wearability requires **unobtrusive placement** (where on body it should go)
 - *areas that are relatively the same size across adults,*
 - *areas that have low movement/flexibility even when the body is in motion, and*
 - *areas that are larger in surface area.*

2) Design for the human body also requires a **humanistic form language** (defining the shape)

- *How to ensure a comfortable, stable fit to dynamic human form?*

3) **Human Movement** (consider the dynamic structure)

- *Consider the many elements that make up any single movement. Elements include the mechanics of joints, the shifting of flesh, and the flexing and extending of muscle and tendons beneath the skin.*

4) **Comfortable attachment** of forms can be created by wrapping the form around the body, rather than using single point fastening systems such as clips or shoulder straps.

5) The **weight** of a wearable should not hinder the body's movement or balance.

6) **Sizing** (for body size diversity)

- *Wearables must be designed to fit as many types of users as possible.*

Unobtrusive Placement

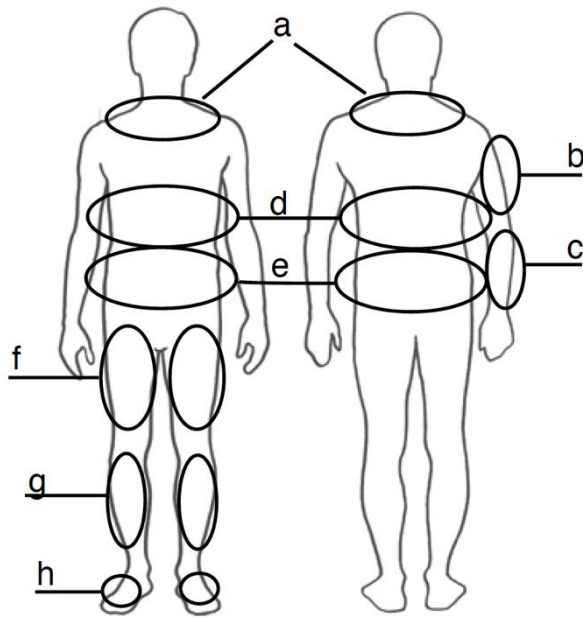
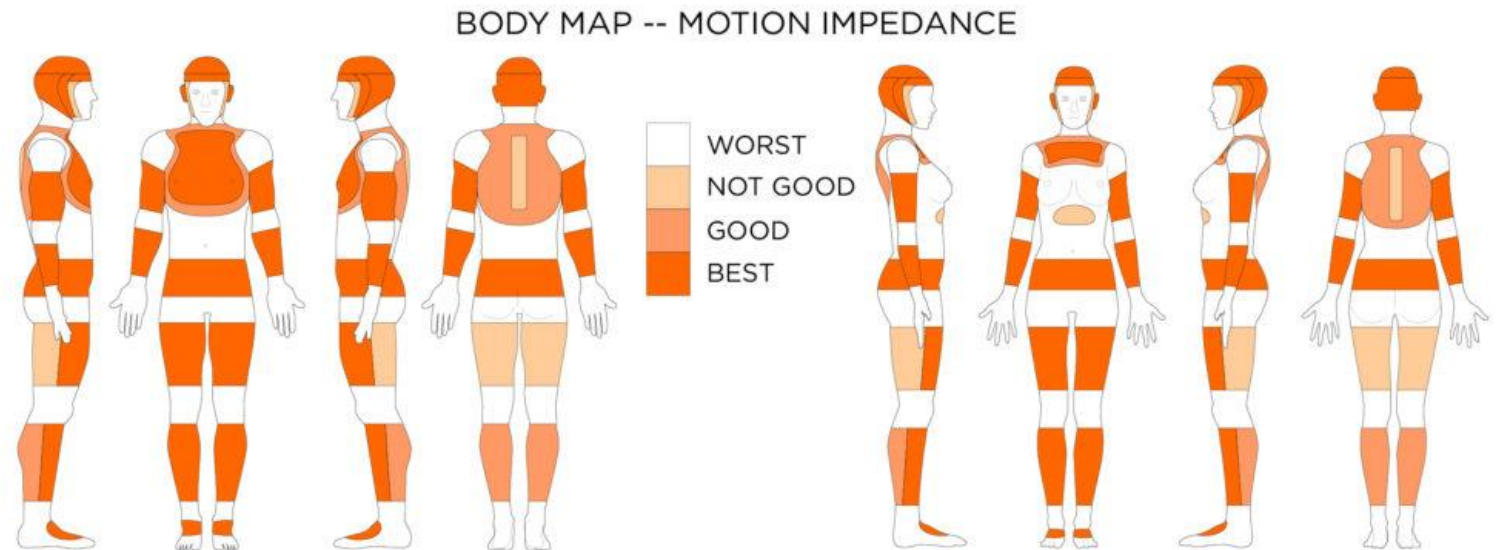


Fig. 1, The general areas we have found to be the most unobtrusive for wearable objects are: (a) collar area, (b) rear of the upper arm, (c) forearm, (d) rear, side, and front ribcage, (e) waist and hips, (f) thigh, (g) shin, and (h) top of the foot.

Source: Gemperle et al. 1998



This body map shows the best places to put wearable devices on the body, where they will be the least obtrusive and cause the least amount of body motion impedance.

Source: <https://www.clintzeagler.com/where-it-body-maps/>

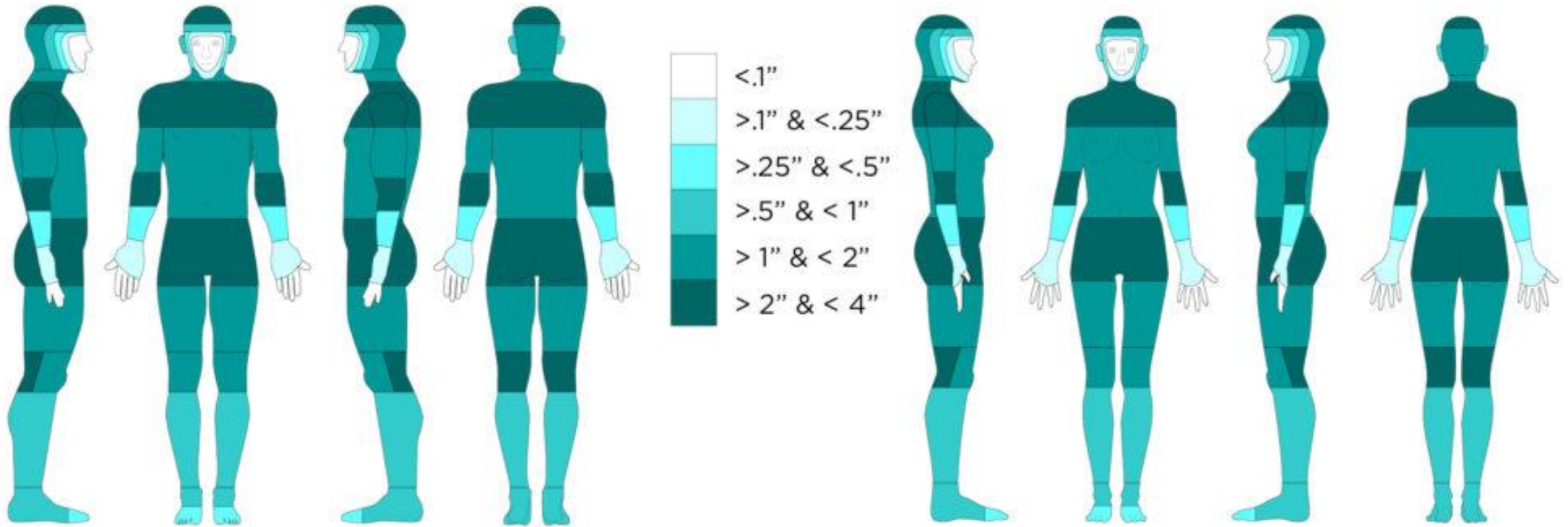
Design Considerations for Body Motion

- Large, bulky, or rigid objects should not be placed on the inside of joints, or the concave areas where the body bends.
- Rigid objects, or flexible but non-elastic objects should not be adhered to the outside of joints in a way that hinders the skin on the outside of the joint from stretching.
- **Smart garments, clothing, or e-textiles should have ample room, or flexible and elastic properties to allow all parts of the body to move effectively.**
- Larger, or rigid objects should be located in zones on the body with relatively limited movement or linkages.

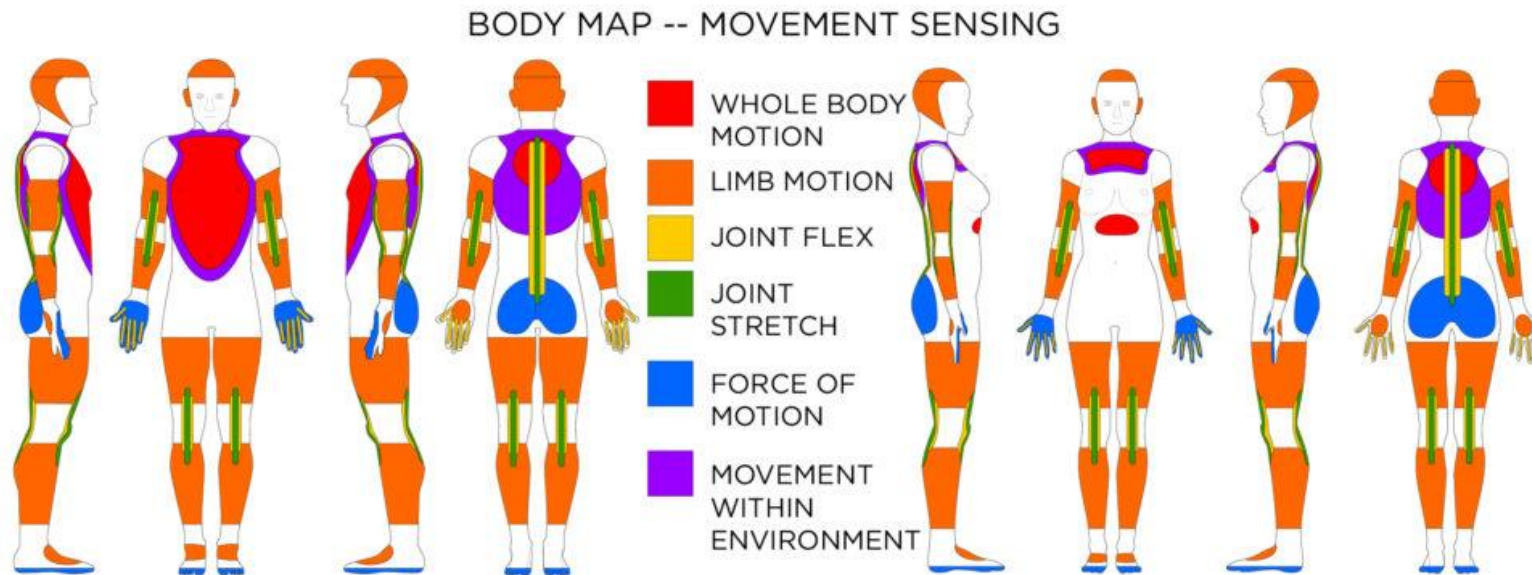
Design Considerations for Weight Distribution

- Weight, load, or the pressure of weight should be placed on the fleshy but non-sensitive parts of the body, avoiding boney areas.
- The lower waist is a good area for heavy loads.
- Weight should be balanced across the body evenly and aligned to the center of gravity if possible.
- Heavy items should not be placed on the body's extremities for long periods of time.
- Batteries for wearable devices tend to be the source of most of the weight. If a device needs a large battery (to last a long time or because it needs large amounts of power to function), place the large battery on the waist. If the wearable needs to be located on a different part of the body for use, then consider distributing the power from the area of use. Finally, consider distributing battery cells instead of using one large battery.

BODY MAP -- PROXEMICS



Proxemics as defined here is a human's perception of self-size. The distance from the body portrayed on this body map indicates how far from the body a wearable device might extend and still be naturally considered part of the person's self-size awareness. Items extending beyond this distance from the body might take a period of time for a person to adjust and account for the object within their personal self-size envelope.



*For sensing whole body motion, and limb motion accelerometers, gyroscopes, and magnetometers can be used at locations indicated. **For sensing joint movement flex or stretch sensors can be used at locations indicated.** Force and explain the impact of movement. If trying to capture movement within an environment, magnetometers (for direction) and barometric pressure sensors (for elevation change) may be used.*

WEAR Scale v2.1 (Likert) (Kelly and Gilbert 2016)

- Focus on **social acceptance of wearables**

1. I think my peers would find this device acceptable to wear.

7. This device seems to be useful and easy to use.

14. This device might restrict movement or physically get in the way. (R)

15. This device is fashionable.

16. This device seems comfortable, not bulky.

20. This device is cool.

22. I can imagine that people would be interested in this device and would not have a problem wearing it.

23. This device is sleek, not clunky.

35. This device could cause health concerns. (R)

25. This device could help people.

26. This device would be generally accepted by the vast majority of people.

27. This device is not weird.

28. The wearer of this device would not be judged negatively by others.

29. This device is similar to existing acceptable devices or accessories.

30. Wearing this device could be considered inappropriate. (R)

31. This device could be considered a normal part of life.

32. Use of this device could hurt the wearer's social reputation. (R)

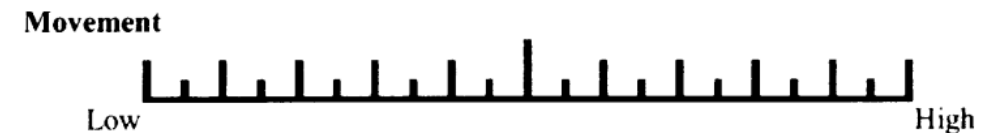
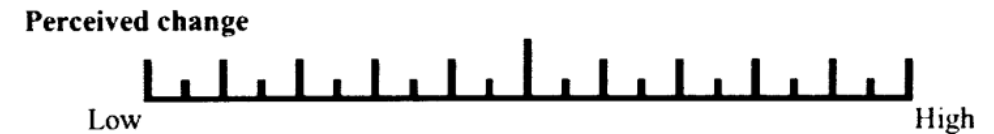
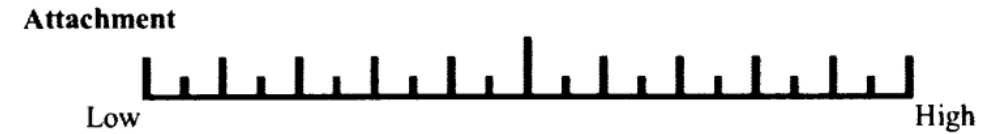
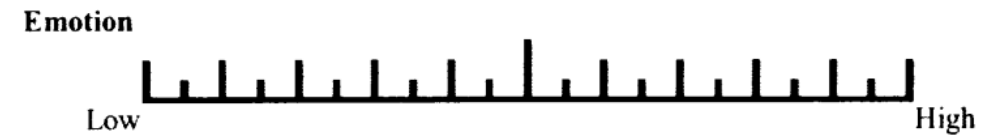
Comfort Rating Scale (CRS) (Knight and Baber 2005)

Source: Knight, J.F.; Baber, C. A Tool to Assess the Comfort of Wearable Computers. Hum. Factors J. Hum. Factors Ergon. Soc. 2005, 47, 77–91

- 21 point scale from 0 (low/good) to 20 (high/bad)

Table 1. Comfort descriptors

Cluster	Description
1	Emotions, concerns about appearance and relaxation
2	Physical feel of the device on the body, attachment
3	Physical effect, damage to the body
4	Feeling physically different, upset
5	The device physically affects movement
6	Worry about the device, safety, and reliability



Title	Endpoints	Description
Emotion	Low/High	I am worried about how I look when I wear this device. I feel tense or on edge because I am wearing the device.
Attachment	Low/High	I can feel the device on my body. I can feel the device moving.
Harm	Low/High	The device is causing me some harm. The device is painful to wear.
Perceived change	Low/High	Wearing the device makes me feel physically different. I feel strange wearing the device.
Movement	Low/High	The device affects the way I move. The device inhibits or restricts my movement.
Anxiety	Low/High	I do not feel secure wearing the device.

Figure 3. Comfort rating scales statements

The CRS scores are often categorized into five levels of wearability to determine if the system needs design changes:

- **WL1 (0–4):** System is comfortable (low impact).
- **WL2 (5–8):** System is wearable, but changes may be necessary (moderate impact).
- **WL3 (9–12):** System is wearable, but changes are advised (large impact).
- **WL4 (13–16):** System is not wearable, uncomfortable (very large impact).
- **WL5 (17–20):** System is not wearable, painful/stressful (extreme impact).

Usability is a measure of how well a specific user in a specific context can use a product/design to achieve a defined goal effectively, efficiently and satisfactorily.

System Usability Scale (Brooke 1996)

The SUS is a 10 item questionnaire with 5 response options.

- 1) I think that I would like to use this system frequently.
- 2) I found the system unnecessarily complex.
- 3) I thought the system was easy to use.
- 4) I think that I would need the support of a technical person to be able to use this system.
- 5) I found the various functions in this system were well integrated.
- 6) I thought there was too much inconsistency in this system.
- 7) I would imagine that most people would learn to use this system very quickly.
- 8) I found the system very cumbersome to use.
- 9) I felt very confident using the system.
- 10) I needed to learn a lot of things before I could get going with this system.

The SUS uses the following response format:

Strongly Disagree 1	2	3	4	Strongly Agree 5
☐	☐	☐	☐	☐

The first step in scoring a SUS is to determine each item's score contribution, which will range from 0 to 4.

- For positively-worded items (odd numbers), the score contribution is the scale position minus 1.
- For negatively-worded items (even numbers), the score contribution is 5 minus the scale position.

To get the overall SUS score, multiply the sum of the item score contributions by 2.5, which produces a score that can range from 0 (very poor perceived usability) to 100 (excellent perceived usability) in 2.5-point increments.

Tips about user tests

- Participation is fully voluntary
- Data collection does not cause damage or harm to participants. Ensure the safety of user tests by implementing measures to prevent incidents, such as falls, during the testing process.
- The real content and purpose of the study are clearly explained to participants beforehand.
- For the project work, do not store participants' personal data such as names, student numbers, contact information, weight, age, etc. Participants can be anonymized and identified using labels like Subject #1, #2, #3.
- In your report, it is enough to tell us how many subjects you have recruited for your user tests and the breakdown by gender.

- In case you are going to include user studies in your thesis work in the future, you can visit this page to see if you need to apply for ethical review of your work.

<https://www.aalto.fi/en/research-art/how-and-when-should-you-apply-for-ethical-review-of-research>

Data processing

- **Descriptive Statistics:** Calculating mean, median, standard deviation, and range
- **Inferential Statistics:** Using T-tests or ANOVA to determine if differences between design versions are statistically significant rather than due to chance.

For this project work, it is good enough if you present descriptive statistics of your user test results.

References

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